Grade 7 Mathematics

OPTIMA ACADEMY ONLINE · COURSE SYLLABUS · 2026–2027

Full Year · Grade 7

FACILITATOR
Optima Academy Online Faculty

EMAIL & OFFICE
Your facilitator's email is posted in Canvas · 239.673.2801 (OAO Front Office)

FLORIDA COURSE
Florida 1205040 · M/J Grade 7 Mathematics · full-year course

OFFICE HOURS
Posted in Canvas · weekly study hall and VR workshops

GRADE LEVEL
Grade 7 (Middle School)

FORMAT
On-Demand (also offered live on the VR campus)

About this syllabus. This is the plan for the course as designed. It is a full-year Grade 7 mathematics course; On-Demand scholars pace through the nine modules across the four quarters of their enrolled year. On-Demand and live-school sections share the same content and expectations; exact pacing, scheduled dates, and a few activities may vary from section to section and term to term. Your facilitator will confirm the specifics for your section.

COURSE OVERVIEW

Grade 7 Mathematics builds the reasoning at the heart of algebra and geometry. Across nine modules, scholars extend arithmetic into the full world of rational numbers and the laws of exponents, then learn to operate on those numbers fluently. From there the course turns abstract: writing and solving algebraic linear expressions, equations, and inequalities; reasoning with percentages and proportions; and analyzing proportional relationships as tables, graphs, and equations. The final modules open into data and space — representing and interpreting statistical data, running simple experiments and reasoning about probability, working with two-dimensional figures and scale, and finding the surface area and volume of cylinders.

The work is hands-on and cumulative. Every lesson pairs a short interactive explanation with worked examples and independent practice, and each module is anchored by a Prior Knowledge Check that activates what a scholar already knows, Mid-Module Checks that catch misunderstandings early, and a Module Summative Assessment that asks the scholar to put it all together. Immersive activities and workshops on Optima's VR campus give scholars a place to see the mathematics and to work through it alongside a facilitator and classmates. The course is aligned to Florida's B.E.S.T. Standards for Mathematics and assumes that fluency and understanding are built together — that reasoning well about a problem depends on real facility with the numbers and operations underneath it.

WHAT SCHOLARS WILL LEARN TO DO

- Rewrite rational numbers in equivalent forms — fractions, decimals, and percents — and apply the laws of exponents.
- Add, subtract, multiply, and divide rational numbers, including negatives, with fluency and understanding.
- Write, interpret, and simplify algebraic linear expressions, and solve multi-step equations and inequalities.
- Recognize and represent proportional relationships as tables, graphs, and equations, and use the constant of proportionality.
- Collect, represent, and interpret statistical data, and describe what a data set does and does not show.
- Design simple experiments and reason about the probability of chance events.

- Reason with percentages — discounts, markups, taxes, tips, and percent change — in real situations.
- Work with two-dimensional figures and scale drawings, and find the surface area and volume of cylinders.

MAJOR ASSESSMENTS & MILESTONES

The course is anchored by a Module Summative Assessment at the end of each module. Target dates below follow a sample full-year pacing; On-Demand scholars pace to their own enrolled term and Live sections follow the class schedule, keeping each milestone roughly in order.

MODULE & SUMMATIVE	COURSE PLACE	SAMPLE TARGET	WHAT IT ASKS
Quarter 1 Summatives Modules 1–3	Q1 · by Oct 9	by Oct 9	Rational numbers & exponents; operations on rational numbers; algebraic expressions, equations & inequalities.
Quarter 2 Summatives Modules 4–5	Q2 · by Dec 18	by Dec 18	Percentages & proportional reasoning; proportional relationships in tables, graphs & equations. *Fall semester ends.*
Quarter 3 Summatives Modules 6–7	Q3 · by Mar 5	by Mar 5	Representing statistical data; simple experiments & probability.
Quarter 4 Summatives Modules 8–9	Q4 · by May 28	by May 28	Two-dimensional figures & scaling; surface area & volume of cylinders.

IMPORTANT DATES — 2026–2027

Aug 12, 2026	First day of school	**Sep 7**	Holiday (Labor Day)
Oct 9	Quarter 1 ends	**Oct 12**	Holiday (Columbus Day)
Nov 11	Holiday (Veterans' Day)	**Nov 23–27**	Thanksgiving break
Dec 18	Quarter 2 ends · fall semester ends	**Dec 21–Jan 1**	Winter break
Jan 5, 2027	Classes resume	**Jan 18**	Holiday (MLK Day)
Feb 15	Holiday (Presidents' Day)	**Mar 5**	Quarter 3 ends
Mar 26–Apr 5	Spring break	**Apr 6**	Classes resume
May 28, 2027	Quarter 4 ends · last day of school	**May 31**	Holiday (Memorial Day)

ASSESSMENT & GRADING

60%
Weekly practice & module checks
Prior Knowledge Checks, Independent Practice, Mid-Module Checks

30%
Module summative assessments
One per module, plus summative reviews

10%
VR campus activities & participation
Immersive activities and workshops

HOW THIS COURSE IS DELIVERED

This course is offered in two formats that share the same content, assignments, and expectations: an **On-Demand (asynchronous)** section you pace yourself, and a **Live (synchronous)** section that meets on a schedule on Optima's VR campus. The two overviews below explain how each one works. Find the one that matches your enrollment.

ON-DEMAND (ASYNCHRONOUS) LEARNING

In the On-Demand section you move through the course at your own pace within each quarter, guided by due dates that keep you on track. All assignments for a quarter open at its start and close at its end. Because the course is On-Demand, there is no grade penalty for missing an individual due date — but any assignment not completed by the end of the quarter receives a zero, and assignments submitted in the final 14 days of a quarter receive a 10% deduction, so steady pacing matters.

Attendance is measured by assignment completion. Per school policy, a scholar who completes no assignment for ten consecutive school days, without notifying the school of an absence in advance, may be dropped from the course. Completion means a sincere attempt — real effort on real work. Rushed work, wrong or corrupted files, irrelevant or nonsense submissions, AI-generated filler, and word-salad responses do not count.

> **You are never on your own.** A live facilitator grades your work, gives feedback, and hosts a weekly study hall and VR workshops.

LIVE (SYNCHRONOUS) LEARNING

In the Live section you join scheduled class meetings on Optima's VR campus, where your teacher leads instruction in real time and you learn alongside classmates. The lessons, assignments, and assessments are the same as the On-Demand course; what differs is the pacing — a Live section moves through the material together on the class schedule rather than at each scholar's own pace, and class time is also used for discussion, questions, and guided practice.

Attendance is measured by presence and participation. You are expected to attend the scheduled live sessions, on time and ready to take part. If you must miss a class, notify the school in advance; your teacher will let you know how to make up the work and any material you missed. As in every Optima course, completing your work means a sincere attempt — real effort on real work — and assignments still open and close with each quarter.

> **You meet regularly with a teacher.** The class gathers throughout the week in VR and on Microsoft Teams for live instruction and discussion.

APPROPRIATE USE OF AI

Optima scholars are preparing for a world shaped by artificial intelligence, and learning to use these tools with good judgment is part of that preparation. You are welcome to use AI tools to support your learning — to explain a concept a different way, check your understanding, or see a similar worked example — unless a particular assignment says otherwise. What AI may never do is the thinking or the work an assignment asks of you: solving the problem, showing the steps, or writing the explanation must be genuinely your own. Submitting AI-generated work as if it were your own is dishonest and, as noted above, counts as an insincere attempt. When you do use an AI tool on an assignment, say so briefly. Above all,

build the underlying skill yourself — the goal is for you to be able to judge whether an AI's answer is any good, which you can only do if you can do the mathematics without it.

LEARNING IN VIRTUAL REALITY

Optima is a virtual-reality school. Alongside your Canvas coursework, you will join immersive sessions on Optima's VR campus, where you meet teachers and fellow scholars, take part in live activities, and work through mathematics together in ways an ordinary classroom cannot. Keep your headset charged and ready for any scheduled VR sessions, and treat the VR campus with the same courtesy and seriousness you would a physical one. If you ever cannot get into VR for a session, tell your facilitator — there is an alternative way to take part so you do not fall behind.

BEING AN OPTIMA SCHOLAR

Optima's mission is an excellent, virtue-centered education, and virtue is at the center of this course as much as any other. Every scholar is expected to practice the seven virtues that shape life at Optima — **courage, courtesy, honesty, perseverance, self-government, service, and responsibility**. In practice that means: do your own honest work and show your reasoning; treat everyone you meet in, through, or because of this course with respect, in writing, in discussion, and in VR; engage actively rather than passively; come to office hours to get help, meet other scholars, and get to know your facilitator; and take responsibility for your own pacing and progress. You belong to a house and a community of scholars, not only to a course.

WHAT YOU NEED & STAYING CONNECTED

- ✓ A computer with reliable internet and access to **Canvas**.
- ✓ A **VR headset**, charged and ready for the immersive activities on Optima's VR campus.
- ✓ A **scientific calculator** and the Grade 7 **reference sheet** (both introduced in the Start Here section).
- ✓ **Microsoft 365** (Word, Excel) with OneDrive, for any documents and data you create and submit.
- ✓ A notebook or binder and pencils, for working problems and keeping your reasoning.

Staying connected. Updates about this course are posted on the course's **Canvas** page; school-wide updates are posted in your Homeroom Canvas course. Check both regularly. Your facilitator holds office hours and is glad to help with the material, to meet you, or to talk through your plans — come by, or reach out any time by email.

Grade 7 Mathematics · Florida course 1205040 (M/J Grade 7 Mathematics) · Full-year middle-school course · Optima Academy Online, 2026–2027. Built from the course content (9 modules across 4 quarters) and the board-approved 2026–2027 academic calendar. Summative target dates reflect a sample full-year pacing; On-Demand scholars pace to their own enrolled term. School calendar dates may be adjusted by the school.